

Project 3: AI Evaluation

Sammi Bronow

AI Modeling in Practice - Summer 2026

AI Use Disclaimer: Claude was used for prompt brainstorming; Gemini was used to enhance slide design

Evaluation Dataset

Dataset Overview

20 cases total, featuring 10 users each paired with 2 contrasting requests.

Each row represents a unique user/request pairing: **10 users x 2 requests each**

Filter Criteria: Users were selected with a minimum of 20 ratings to ensure a meaningful CF signal.

Candidate Generation

Top 10 candidates per user were generated via the Project 2 CF model (IBCF, pearson_baseline, k=20).

Passed to Re-ranker:

- Title
- Author
- Publication Year
- Average Rating

Core Requests

Applied to Every User

Request 1 (Mood/Tone)

"Something dark and scary that I can't put down"

Request 2 (Content/Plot)

"Something with an intense, fast-paced plot and a shocking ending"

Prompt Design

How Prompt B differs from Prompt A:

- 01. Context:** Explicitly states candidates were pre-selected by a CF model, setting clear expectations and context.
- 02. Ranking Rule:** Instructs the model to prioritize thematic fit over raw popularity or average rating.
- 03. Structured Criteria:** Enforces three distinct evaluation criteria applied in a strict, defined order.
- 04. Explanation Standard:** Requires justifications deeply grounded in content and style rather than brief summaries.

*Both prompts used **Gemini 2.5 Flash** and **temp = 0**.*

*Prompt B was **iteratively refined** in the Braintrust playground.*

Prompt A — Baseline

You are a book recommender. From the candidate list, pick and rank the top 3 books that best match the reader's preference. For each pick, provide a short reason. Use the exact titles from the candidate list. Do not recommend books outside of the candidate list.

Prompt B — Designed

You are an expert book concierge. You are given a curated list of candidates pre-selected by a collaborative filtering model for this specific reader. Re-rank and select the top 3 books that best match the reader's stated preference. Ranking order is very important. Prioritize clear thematic matches over average rating; that is, a highly-rated book that is not a clear match to the user's request should rank below a lower-rated book that is a clear match.

Apply these criteria in order:

1. Thematic or tonal fit to the stated request
2. Quality signal (average rating)
3. Specificity (prefer books where the match is concrete and explainable)

For each pick, write an explanation that is grounded in the book's content or style and clearly connects to the reader's request. Recommend only from the candidate list, using exact titles and authors.

Scorers

Ranking Judge

Purpose:

Evaluates whether the top 3 picks fit the reader's request and are well-ordered.

Constraint:

Penalizes picks outside the candidate list.

Explanation Judge

Purpose:

Evaluates whether each explanation is specific and grounded in the book's metadata (title, author, year, rating) and clearly connected to the reader's request.

Constraint:

Penalizes vague or generic reasoning.

Model & Scale

Model: OpenAI GPT-4o-mini

Scale: great = 1.0, ok = 0.5, poor = 0

Why GPT-4o-mini?

Used a different model family from the generator (which used Gemini 2.5-Flash) to **reduce self-serving bias**.

See Appendix A for screenshots of actual scorer user and system messages.

Results

Scores and metrics		Prompt A Base	Prompt B Comparison
🔍 Comparison grade			Tradeoff Base has better latency and load. Comparison has better ExplanationJudge and RankingJudge.
% ExplanationJudge	92.5% AVG	↘ 3	100% -7% ↘ 3
% RankingJudge	52.5% AVG	↘ 12	70% -17% ↘ 7
🕒 Duration	0.997s SUM	↗ 20 ↘ 7	1.09s +0.098s ↗ 12 ↘ 7

**Prompt B
achieved
100%
explanation
fit!**

+17.5% Ranking Fit Gain

Driven by explicit instruction to prioritize thematic match over average rating.

+7.5% Explanation Quality

Achieved via the strict requirement to ground each explanation in book metadata.

Ranking Fit Plateaued at 70%

Multiple iterations of Prompt B, which included **explicit non-fiction constraints** and **temperature adjustments** (0.3 vs. 0), had a **negative effect** on ranking fit.*

*See Appendix B for more detail.

Trace Case Study

User Request #2: “something with an intense, fast-paced plot and a shocking ending”

Trace #1: Prompt B helped

Prompt B improved both judges' scores from **50% to 100%**. Prompt B's explicit instruction to ground explanations in plot details surfaced a better third pick in terms of genre (*Beautiful Disaster* over *Marked*), and its grounded explanation requirement forced more specific plot-level reasoning. This example shows that **when the candidate pool contains strong matches, Prompt B's mechanisms work as designed.**

	Name	Input	Output	Ex...	Tags	Cl...	% ExplanationJudge	% RankingJudge
							92.5% AVG 100% -7%	52.5% AVG 70% -17%
4	★	eval	{ "candidates_text": "- 'If I Stay (If I Stay, #1)' by Gayle Forman (2009, avg rating 3.96) - 'Beautiful...	• Here ar... • Here ar...	-	-	• 50% • 100.0% -50.0%	• 50% • 100.0% -50.0%

Trace #2: Prompt B did *not* help

Both prompts scored **identically** and landed on the same third pick, *Kidnapped*, flagged by the Ranking Judge as a weak fit due to its lack of shocking ending. Notably, the pool included *E is for Evidence*, a mystery novel with plot twists, which went unpicked. Since the user request does not define what “intense” means, the model's read (adventure vs. emotional shock) was defensible but scored poorly. Conversely, both prompts earned 100% explanation ratings here, emphasizing the importance of **independent judges**: each one identifies problems that the other might miss.

	Name	Input	Output	Ex...	Tags	Cl...	% ExplanationJudge	% RankingJudge
							92.5% AVG 100% -7%	52.5% AVG 70% -17%
5	★	eval	{ "candidates_text": "- 'The Adventures of Tom Sawyer & Adventures of Huckleberry Finn' by Mark Twain (1876...	• Here a... • Here a...	-	-	• 100% • 100.0%	• 50% • 50.0%

Shipping Recommendation

Prompt B wins on both scoring mechanisms.

92.5% → **100%**
Explanation Quality

52.5% → **70%**
Ranking Fit

Tradeoff: Precision vs. Recall

Weak matches (false positives) erode trust, while missing strong matches (false negatives) cost engagement. Ultimately, which matters more is a business call.

Recommendation: Optimize Precision

For a recommendation app, repeat engagement relies on user trust. Prompt B prioritizes thematic fit over rating, so weak matches rank lower, effectively favoring precision over recall. It does so with minimal latency cost, though **token usage is 74% higher** due to prompt length.* This is a meaningful difference at scale, but likely a small price relative to the cost of a lost user.

Next Steps:

- Increase k beyond 20 to widen the candidate pool, surfacing more potential thematic matches for user requests. Ideally, this will address the ranking fit plateau.
- Expand the evaluation dataset to cover request types beyond mood and plot (e.g., narrative style, literary devices, etc.)

*See Appendix B for exact token usage and latency figures across prompts.

Evaluation Design: Limitations & Alternatives

Limitations of This Design

- **Sample size:** 10 readers x 2 requests = 20 cases. Patterns are visible, but results could shift with more data.
- **No ground truth:** No human-verified golden dataset was created to validate judge scores.
- **Narrow request scope:** Only mood/tone and plot-based requests were tested. Other literary devices were not explored.
- **Judge bias:** Using a different LLM for scoring mitigated self-preference bias, but position bias and length bias were not explicitly tested.

Alternative Evaluation Options

- **Offline metrics:** Precision@k and NDCG can be used to measure ranking quality against held-out ratings.
- **Human evaluation:** Held-out reader ratings can be used to form a golden dataset to calibrate LLM judge scores.
- **Behavioral/implicit signals:** Clicks, favorites, or follow-up requests measure task completion, which is a stronger signal than turn-level explanation quality.
- **A/B testing in production:** Ship both prompts to real users and measure which drives more engagement.

Appendix A: Screenshots of Eval Dataset, Prompts, and Scorers

Project 3 Eval Data V2

	Import	+ Row	All dataset rows view	Reset	Save as	Q Add filter
	Created	Input				
1	Jun 25 04:49:41 PM	{"candidates_text":{"The Hunger Games (The Hunger Games, #1)" by Suzanne ...				
2	Jun 25 04:49:41 PM	{"candidates_text":{"The Hunger Games (The Hunger Games, #1)" by Suzanne ...				
3	Jun 25 04:49:41 PM	{"candidates_text":{"Crescendo (Hush, Hush, #2)" by Becca Fitzpatrick (20...				
4	Jun 25 04:49:41 PM	{"candidates_text":{"Crescendo (Hush, Hush, #2)" by Becca Fitzpatrick (20...				
5	Jun 25 04:49:41 PM	{"candidates_text":{"Lock and Key" by Sarah Dessen (2008, avg rating 4.01...				
6	Jun 25 04:49:41 PM	{"candidates_text":{"Lock and Key" by Sarah Dessen (2008, avg rating 4.01...				
7	Jun 25 04:49:41 PM	{"candidates_text":{"2001: A Space Odyssey (Space Odyssey, #1)" by Arthur...				
8	Jun 25 04:49:41 PM	{"candidates_text":{"2001: A Space Odyssey (Space Odyssey, #1)" by Arthur...				
9	Jun 25 04:49:41 PM	{"candidates_text":{"I Am Malala: The Story of the Girl Who Stood Up for ...				
10	Jun 25 04:49:41 PM	{"candidates_text":{"I Am Malala: The Story of the Girl Who Stood Up for ...				
11	Jun 25 04:49:41 PM	{"candidates_text":{"The Woman in Cabin 10" by Ruth Ware (2016, avg ratin...				
12	Jun 25 04:49:41 PM	{"candidates_text":{"The Woman in Cabin 10" by Ruth Ware (2016, avg ratin...				
13	Jun 25 04:49:41 PM	{"candidates_text":{"Johnny Got His Gun" by Dalton Trumbo (1939, avg rati...				
14	Jun 25 04:49:41 PM	{"candidates_text":{"Johnny Got His Gun" by Dalton Trumbo (1939, avg rati...				
15	Jun 25 04:49:41 PM	{"candidates_text":{"The Adventures of Tom Sawyer & Adventures of Huckleb...				
16	Jun 25 04:49:41 PM	{"candidates_text":{"The Adventures of Tom Sawyer & Adventures of Huckleb...				
17	Jun 25 04:49:41 PM	{"candidates_text":{"If I Stay (If I Stay, #1)" by Gayle Forman (2009, av...				
18	Jun 25 04:49:41 PM	{"candidates_text":{"If I Stay (If I Stay, #1)" by Gayle Forman (2009, av...				
19	Jun 25 04:49:41 PM	{"candidates_text":{"2018: Odyssey Two (Space Odyssey, #2)" by Arthur C. ...				
20	Jun 25 04:49:41 PM	{"candidates_text":{"2018: Odyssey Two (Space Odyssey, #2)" by Arthur C. ...				

Eval dataset

Name

Slug

Prompt A

prompt-a-5e89

Model

Gemini 2.5 Flash

Params

Messages

User

User request: {{input.user_request}}
Book candidates: {{input.candidates_text}}

+ Message part

System

You are a book recommender. From the candidate list, pick and rank the top 3 books that best match the reader's preference. For each pick, provide a short reason. Use the exact titles from the candidate list. Do not recommend books outside of the candidate list.

Mustache

T Text output

+ Tool/MCP

+ Message

Prompts

Name

Slug

RankingJudge

rankingjudge-bb36

Model

GPT-4o mini

Params

Messages

User

User request: {{input.user_request}}
Candidates: {{input.candidates_text}}
Model's ranking: {{output}}

+ Message part

System

You are a judge evaluating whether the top 3 book recommendations fit the reader's request. Score how well the top 3 picks, and their order, fit the user's request. If a returned title is not from the candidate list, grade poor. For each of the 3 picks, consider whether it fits the request. Then give an overall grade based on the set.

great = top 3 clearly fit and are well-ordered
ok = top 3 roughly fit, but the order or one pick is a weak fit
poor = the picks contradict the request, or a title is invented

Mustache

+ Message

Scorers

Name

Slug

ExplanationJudge

explanationjudge-1384

Model

GPT-4o mini

Params

Messages

User

User request: {{input.user_request}}
Candidates: {{input.candidates_text}}
Model's ranking: {{output}}

+ Message part

System

You are a judge evaluating the quality of explanations given for book recommendations. For each of the 3 picks, consider whether the explanation is specific and grounded in the book's metadata (title, author, year, rating), and clearly connects to the reader's request. Penalize vague or generic explanations like "this book is a great fit" with no supporting detail.

great = all 3 explanations are specific and clearly tied to the request
ok = explanations are partially specific, or one is vague
poor = explanations are generic, repetitive, or disconnected from the request

Output type

% Score

This score will output a number score (0-1)

Choice scores

Choice scores are required when using LLM judge scorers. The model will be forced to choose one of the choices using a tool schema. All choices and scores must be unique.

Choice

Score (0 to 1)

ok

0.5

poor

0

great

1

Add choice score

Use chain of thought (CoT)

Appendix B: Screenshot of Results Comparison

Comparisons

Prompt B

Prompt B v2

Prompt B v3

Prompt

Prompt A

Dataset

Project 3 Eval Data V2

Parameters

None

Scorers and distribution

% ExplanationJudge

0%

50%

100%

% RankingJudge

0%

50%

100%

Metadata

YAML

1

dataset: Project 3 Eval Data V2

2

model: gemini-2.5-flash

3

temperature: 0

4

use_cache: true

Tags

As noted in Slide 5 (“Results”), two additional iterations of Prompt B were tested in an attempt to increase the Ranking Judge score from 70%. Prompt B v2 maintained the same system message but **increased temperature to 0.3**. Prompt B v3 **reverted temperature to zero and added one line to system message**: “When evaluating fit, deprioritize non-fiction, memoirs, and short-story collections unless the reader explicitly welcomes them.” Both iterations beat the baseline ranking score, but **failed to improve upon the ranking score of the original Prompt B (70%)**.